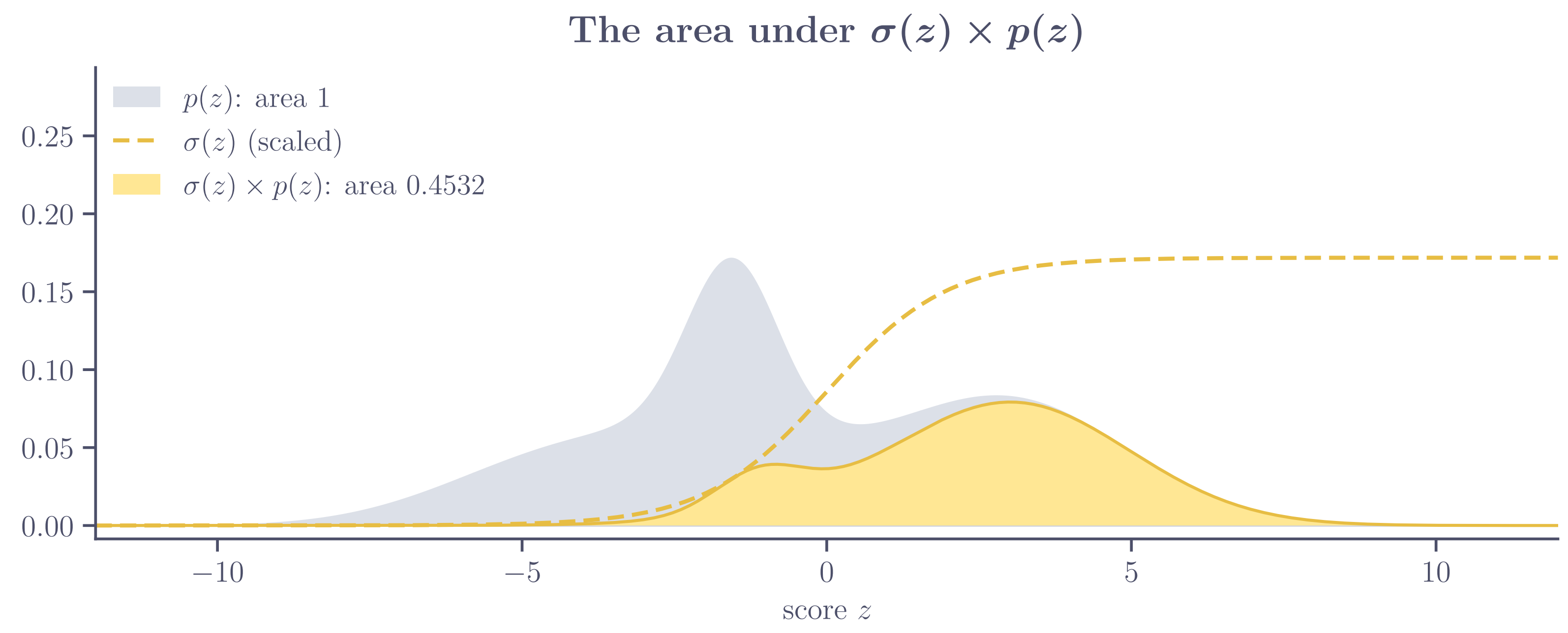
GOAL

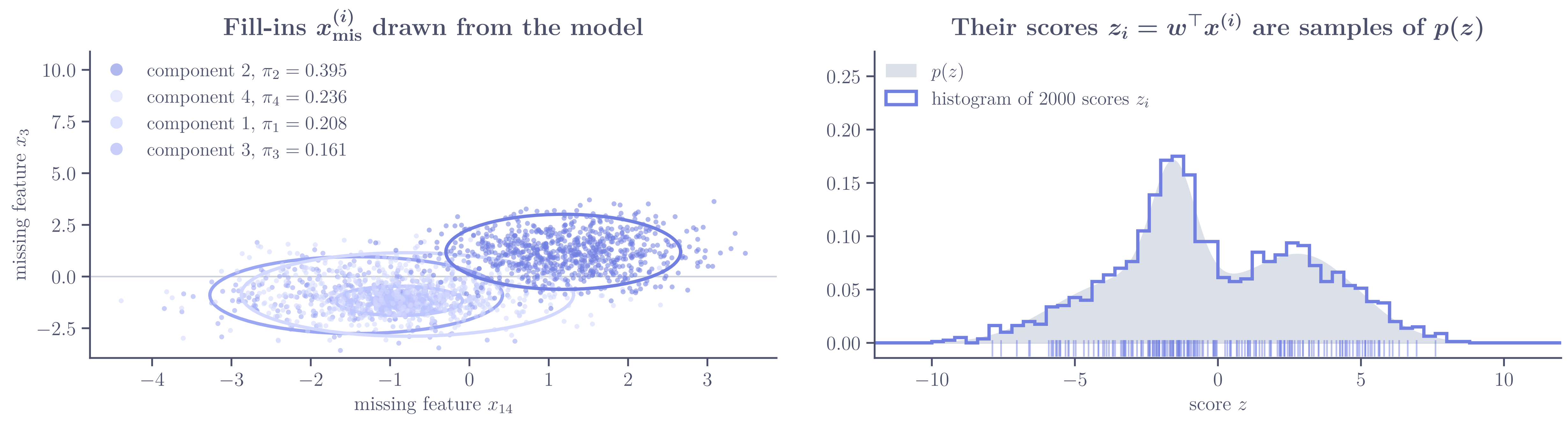
What we want to compute


The area under $\sigma(z) \times p(z)$

Some features are missing, so the score $z = w^\top x$ is uncertain: $p(z)$, from the density model (grey, area 1). The expected prediction $\mathbb{E}[\sigma(z) \mid x_{\text{obs}}]$ is the yellow area under $\sigma(z) \times p(z)$. Monte Carlo does not compute it at all, it estimates it by trying out random fill-ins of the missing features.

STEP 1

Fill in the missing features at random, from the model

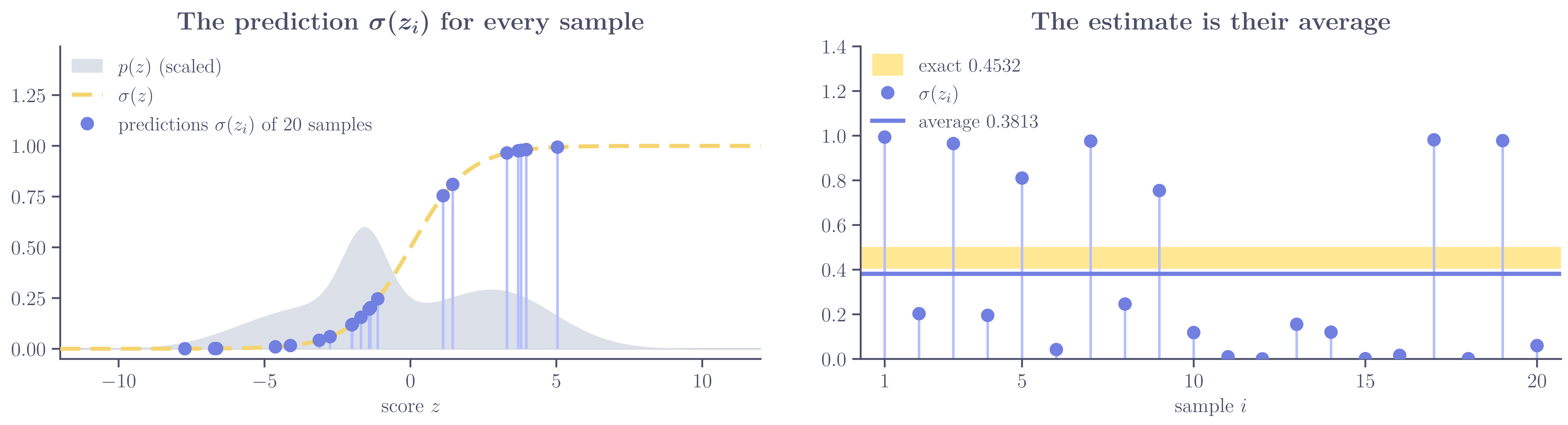

Fill-ins $x_{\text{mis}}^{(i)}$ drawn from the model

Their scores $z_i = w^\top x^{(i)}$ are samples of $p(z)$

Draw n fill-ins $x_{\text{mis}}^{(i)} \sim p(x_{\text{mis}} \mid x_{\text{obs}})$ from the density model: pick a component k with weight $\pi_k = p(k \mid x_{\text{obs}})$, then the missing features from its Gaussian (dots, coloured by component, ellipses 2 standard deviations). Shown are 2 of the 9 missing features. Every fill-in, together with the observed features, gives one score $z_i = w^\top x^{(i)}$. The scores are samples from $p(z)$: their histogram (indigo) follows the density (grey). The model only has to be able to sample, nothing about $p(z)$ is computed.

STEP 2

Predict for every fill-in and average

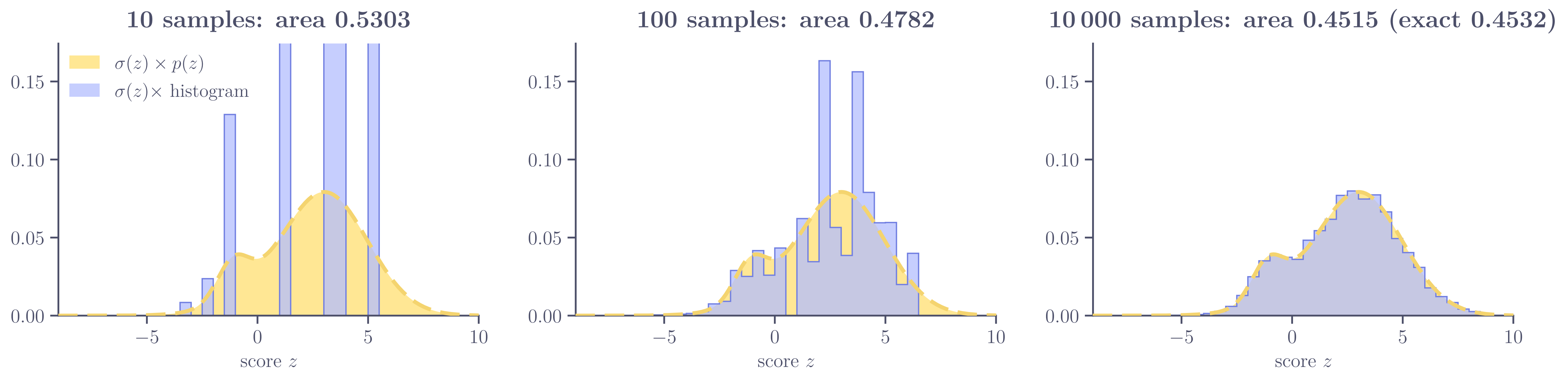

The prediction $\sigma(z_i)$ for every sample

The estimate is their average

The classifier is run on every completed input, which gives the prediction $\sigma(z_i)$ (dots on the yellow $\sigma(z)$). Where $p(z)$ has more mass, more samples land, so the predictions there count more often. The estimate is the plain average $\frac{1}{n}\sum_i \sigma(z_i)$ of the predictions (indigo line), here of the 20 samples on the left, against the exact value (yellow). Every sample counts the same, the weighting by $p(z)$ comes from where the samples are.

STEP 3

The average is an area under $\sigma(z)\times$ a histogram


10 samples: area 0.5303

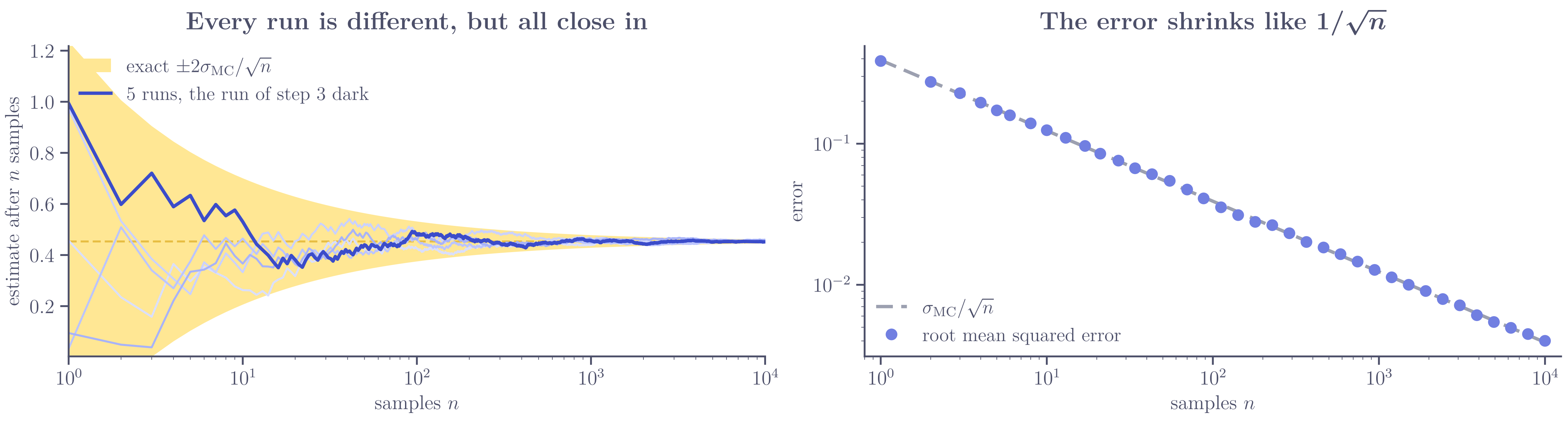
100 samples: area 0.4782

10 000 samples: area 0.4515 (exact 0.4532)

Every panel: $\sigma(z)$ times the histogram of the first n samples (indigo bars, each sample adds $\sigma(z_i)/n$ to the area of its bar), against $\sigma(z) \times p(z)$ (yellow, dashed) and its area. The area of the bars is exactly the Monte Carlo estimate. With few samples the bars are a rough, random picture of the goal, with many they fill it out.

STEP 4

The estimate is random, its error shrinks slowly


Every run is different, but all close in

The error shrinks like $1/\sqrt{n}$

Every run of Monte Carlo gives a different estimate (indigo lines: 5 runs, the dark one is the run of step 3). They all close in on the exact value, inside a band of $\pm 2\sigma_{\text{MC}}/\sqrt{n}$ around it (yellow), where $\sigma_{\text{MC}}^2 = \text{Var}[\sigma(z)] = 0.1529$ is the variance of one prediction. The root mean squared error over 500 runs falls like $\sigma_{\text{MC}}/\sqrt{n}$ (dashed): every digit of accuracy costs 100 times more samples. Since $\sigma(z) \in [0, 1]$ the variance is at most 1/4, whatever the model, but the error never becomes 0. The estimate at 10 000 samples is 0.4516 (exact 0.4532).